



Mathematics Part 1 — Cbse Periodic Test (Pt) — 40 Marks

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	Which identity can be used to expand the product of two binomials of the form $(ax + b)(cx + d)$? (A) $(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$ (B) $(ax + b)(cx + d) = acx^2 + (ab + cd)x + bd$ (C) $(ax + b)(cx + d) = acx^2 + (ad - bc)x + bd$ (D) $(ax + b)(cx + d) = acx^2 + (ad + bc)x - bd$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
2	The centre of the unique circle passing through three non-collinear points is called the: (A) Incentre (B) Circumcentre (C) Orthocentre (D) Centroid <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
3	One of the columns on the Ishango bone groups notches into which of the following sets of numbers? (A) 2, 4, 8, 16 (B) 11, 13, 17, 19 (C) 1, 3, 5, 7 (D) 10, 12, 14, 16 <i>[Ch 3: The World of Numbers]</i>	1



Q.NO.	QUESTIONS	MARKS
4	The algebraic expression $36x^2 + 12x + 1$ is written in the form $a^2 + 2ab + b^2$ to factorise it using the identity $(a + b)^2 = a^2 + 2ab + b^2$. What are the values of a and b in this case? (A) $a = 36x$, $b = 1$ (B) $a = 6x$, $b = 1$ (C) $a = 6$, $b = 1$ (D) $a = 6x$, $b = 12$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
5	Which term of the sequence 2, $2\sqrt{2}$, 4, ... is 128? (A) 14th (B) 15th (C) 16th (D) 13th <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
6	What is the side of a cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$ cubic units? (A) $p + q$ (B) $p + 2q$ (C) $2p + q$ (D) $p + 8q$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
7	Assertion (A): In an arithmetic progression, each term after the first is obtained by adding a fixed number to the previous term. Reason (R): The fixed number added in an arithmetic progression is called the common difference. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	1
8	Assertion (A): The expression $10x - x^2$ is a linear function. Reason (R): The expression $2x + 3$ is a linear function. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Introduction to Linear Polynomials]</i>	1

Section B

Attempt all 4 questions. Each carries 2 mark(s).



Q.NO.	QUESTIONS	MARKS
9	A pattern of square tiles starts with 1 tile in Stage 1, 3 tiles in Stage 2, and 5 tiles in Stage 3. Write the linear expression for the number of tiles at Stage n. <i>[Ch 2: Introduction to Linear Polynomials]</i>	2
10	How many circles can be drawn passing through three non-collinear points? <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	2
11	A chord of length 24 cm is drawn in a circle of diameter 26 cm. Find the distance from the centre of the circle to the chord. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	2
12	Are rational numbers closed under multiplication? Justify your answer. <i>[Ch 3: The World of Numbers]</i>	2
Section C Attempt 3 out of 4.		
13	The first term of an AP is 11 and the common difference is -3. Find the 10th term of this AP. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	3
	OR In a circle, a chord is 5 cm away from the centre. If the radius of the circle is 13 cm, what is the length of the chord? <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
14	For a linear equation of the form $y = ax$ where $a > 0$, how does the steepness of the straight line change when $a > 1$ compared to when $a < 1$? Where do all such lines pass through? <i>[Ch 2: Introduction to Linear Polynomials]</i>	3
15	Write the identity for the product of $(x + y)$ and $(x^2 - xy + y^2)$. <i>[Ch 4: Exploring Algebraic Identities]</i>	3
Section D Attempt 2 out of 3.		
16	Find the sum of the consecutive integers from 25 to 58 by using the formula for the sum of the first n natural numbers. Show the step-by-step calculation. <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	5
	OR State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$. <i>[Ch 4: Exploring Algebraic Identities]</i>	



Q.NO.	QUESTIONS	MARKS
17	The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in the circle. Find the distance of the chord from the centre. Also, a cyclic quadrilateral ABCD is inscribed in this circle with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find the measures of $\angle C$ and $\angle D$. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	5
Section E Attempt 1 out of 1.		
18	Case Study: Book Club Subscriptions and Linear Polynomials A book club charges a joining fee of ₹100 and ₹20 for every book borrowed. The total amount a member pays is represented by the linear polynomial $P(b) = 20b + 100$, where b is the number of books. The amount increases by a constant value for each additional book borrowed, representing linear growth. Another club charges no joining fee but ₹40 per book, represented by the polynomial $Q(b) = 40b$. Both $P(b)$ and $Q(b)$ are examples of univariate polynomials. Polynomials can also be thought of as input-output processes, where substituting a value for the variable gives an output. (i) What is the degree of the polynomial $P(b) = 20b + 100$? (ii) Identify the constant term of the polynomial $P(b) = 20b + 100$. (iii) Polynomials can be thought of as input-output processes. Calculate the total amount a member of the first club will pay if they borrow 5 books. (iv) Does the first club's plan represent linear growth or linear decay? Justify your answer based on the slope of the line. <i>[Ch 2: Introduction to Linear Polynomials]</i>	5

